PitchShifter

Ticco Ross · User manual · version 1.0.3

Overview

PitchShifter is a three-voice harmonizer. It adds up to three pitch-shifted copies of your signal next to the original, each with its own level, pan, tuning and delay with feedback. Pitch shifting is done with modulated delay lines, which gives it a smooth, slightly animated character that works on vocals, synths, guitars and drums alike.

Formats: VST3 and standalone on macOS, Windows and Linux, plus Audio Unit on macOS. Mono and stereo inputs and outputs are supported.

Installation

macOS	Open the .pkg and choose the components to install (Audio Unit, VST3, standalone app). The installer is not signed by Apple: the first time, right-click it and choose Open.
Windows	Run the Setup .exe. The VST3 goes to C:\Program Files\Common Files\VST3. A zip with the same files is also available.
Linux	Unpack the .tar.gz and run install.sh. As a normal user it installs to ~/.vst3 and ~/.local; with sudo it installs system-wide.

Controls

Input Voice — the unprocessed signal.

Mute	Silences the original signal so only the shifted voices are heard.
Gain	Level of the original signal, -48 to +12 dB. The meter beside it shows its level.
Pan	Stereo position, 0 (left) to 100 (right); 50 is centre.
Dry/Wet	Balance between the untouched input and the processed mix, 0 to 100 %.

Voice 1, 2 and 3 — each voice has the same controls.

Mute	Turns the voice off.
Gain	Voice level, -48 to +12 dB, with its own meter.
Pan	Stereo position of the voice, 0 to 100.
St	Transposition in semitones, -12 to +12. Use + and - to step.
Cent	Fine tuning, about -1 to +1 semitone. Small values make a chorus-like detune.
Delay	Delays the voice by 0 to 5 seconds, for echoes and arpeggiated harmonies.
Fb	Feedback of the voice's delay, 0 to 1. Higher values repeat the shifted echo; with a pitch interval set, each repeat climbs or falls further.

Top bar

Preset menu	Shows the current preset. Click it to pick a factory or user preset, or to Save As, Import, Delete the current user preset, or open the preset folder.
■ / ■	Step to the previous or next preset.
Import icon	Load a preset file (.xml) from anywhere on your computer; it is copied into your preset folder.
Save icon	Save the current settings as a user preset.
Undo / Redo	Step back or forward through parameter changes.

Presets

Five factory presets are built in: **Maj Seventh**, **Min Seventh** and **Dim Seventh** (seventh-chord harmonies), **Mystico** and **Delirium**. Your own presets appear under **User** in the preset menu. They are saved as .xml files in:

macOS	~/Library/Audio/Presets/Ticco Ross/PitchShifter
Windows	Documents\Ticco Ross\PitchShifter\Presets
Linux	~/.local/share/Ticco Ross/PitchShifter/Presets

The current preset name is saved with your DAW project. You can copy preset files between computers; drop them in the folder or use Import.

Tips

• Classic harmony: Voice 1 at +4 St, Voice 2 at +7 St, panned left and right.
• Thickening: one voice at +0.1 Cent and one at -0.1 Cent, delays of a few milliseconds.
• Shimmer: a voice at +12 St with 0.3 s delay and feedback around 0.5.
• Set Dry/Wet to 100 and mute the input voice to hear only the harmonies.

PitchShifter is free software released under the GNU GPL v3. Source code: github.com/Spunkmeyer77/PitchShifter